

Beyond Pixel Fidelity: Minimizing Perceptual Distortion and Color Bias in Night Photography Rendering

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pHVI-ISPNet renders night-time RAW into RGB in a color-and-intensity decoupled space, and is supervised on how the result looks rather than on pixel error alone.

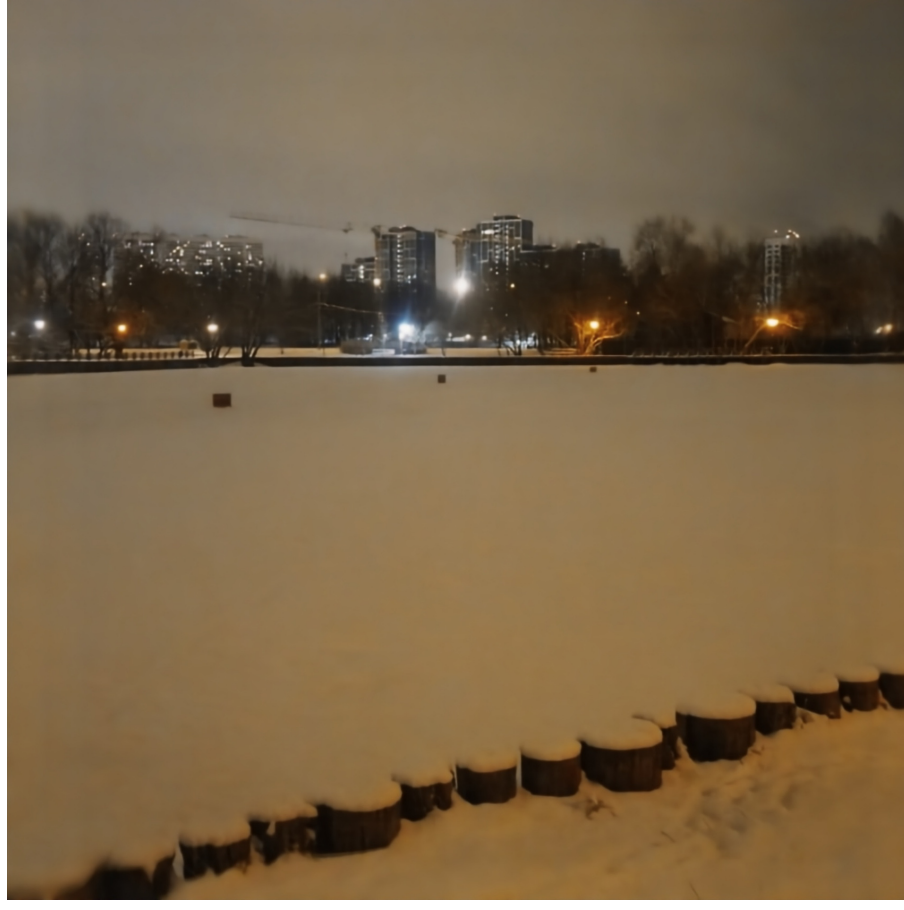
1 The problem

why a high PSNR can still look wrong

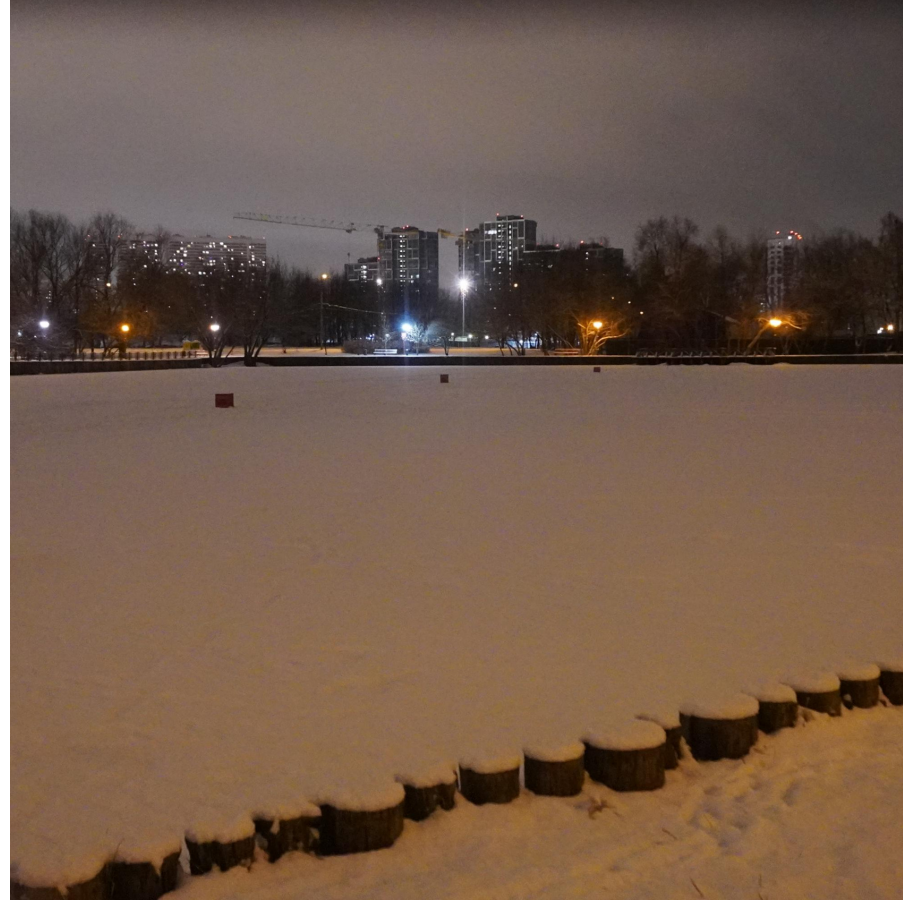
- A night frame holds **near-black regions and bright point lights at once**, lit by several spectrally different sources, under heavy sensor noise.
- PSNR and SSIM score average pixel agreement, so they **barely register a global color cast** or a flattened tone curve.
- In the NTIRE 2025 challenge on night photography rendering, the **objective winner and the human-preference winner were different methods**.

Fidelity leader ≠ perceptual leader

Tuning for distortion metrics leaves visible color bias and tone artifacts on the table. We optimize the perceptual side and keep the fidelity.



Highest PSNR in the challenge
NJUST_KMG, 23.82 dB



Ground truth
Reference rendering

The snow gives it away: best fidelity score in the challenge, yet the field takes on a yellow-green tint – ΔE 5.85, LPIPS 0.433.

Where we land

5.42

ΔE_{2000} color
best of 12, from 5.85

0.388

LPIPS
best of 12, from 0.400

23.63

PSNR, dB
2nd of 12, 0.19 behind

0.785

SSIM
2nd of 12, 0.008 behind

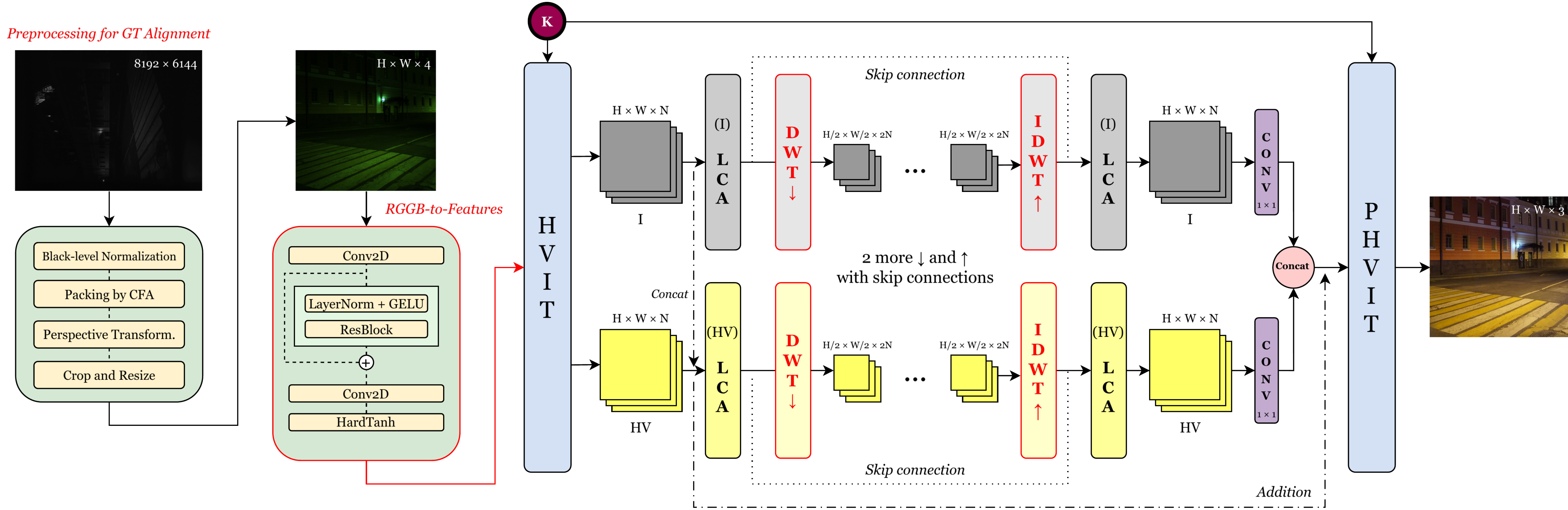
NTIRE 2025 NPR test set, 200 images at 2000 × 2000 px.

2 How it works

CIDNet base, four RAW-domain changes

The HVI transform splits the image into a **color stream (HV)** and an **intensity stream (I)**, so tone mapping cannot drag the color with it. Two dual-branch U-Nets process the streams separately and fuse at the end; four changes adapt that base to RAW night data.

$L_{\text{total}} = L_p + \lambda_{\Delta E} L_{\Delta E} + \lambda_{\text{FDM}} L_{\text{FDM}}$ L_p holds six reconstruction terms – L1, SSIM and edge, in both RGB and HVI – each scaled per sample by $\alpha = \max(\mu_{\text{GT}}/\mu_{\text{pred}}, \mu_{\text{pred}}/\mu_{\text{GT}})$. Card gains below are incremental over the HVI-CIDNet base.



Packed RGGB in, rendered RGB out. Red outlines mark our additions: the RGGB-to-Features stem and the DWT/IDWT resampling.

RGGB-to-Features stem

Takes the packed Bayer frame directly and learns demosaicking and noise suppression as features, instead of inheriting a fixed ISP.

ΔE 7.06 → 5.61
PSNR 21.71 → 23.48

Wavelet feature propagation

DWT and IDWT replace pooling and striding. All four sub-bands are kept and fused 1×1, so fine detail survives the depth.

LPIPS 0.409 → 0.399
 ΔE 5.61 → 5.51

Feature-distribution loss

Exact feature distribution matching aligns the sorted deep features of output and target, pinning down color rather than per-pixel error.

LPIPS 0.399 → 0.379
 ΔE 5.51 → 5.53

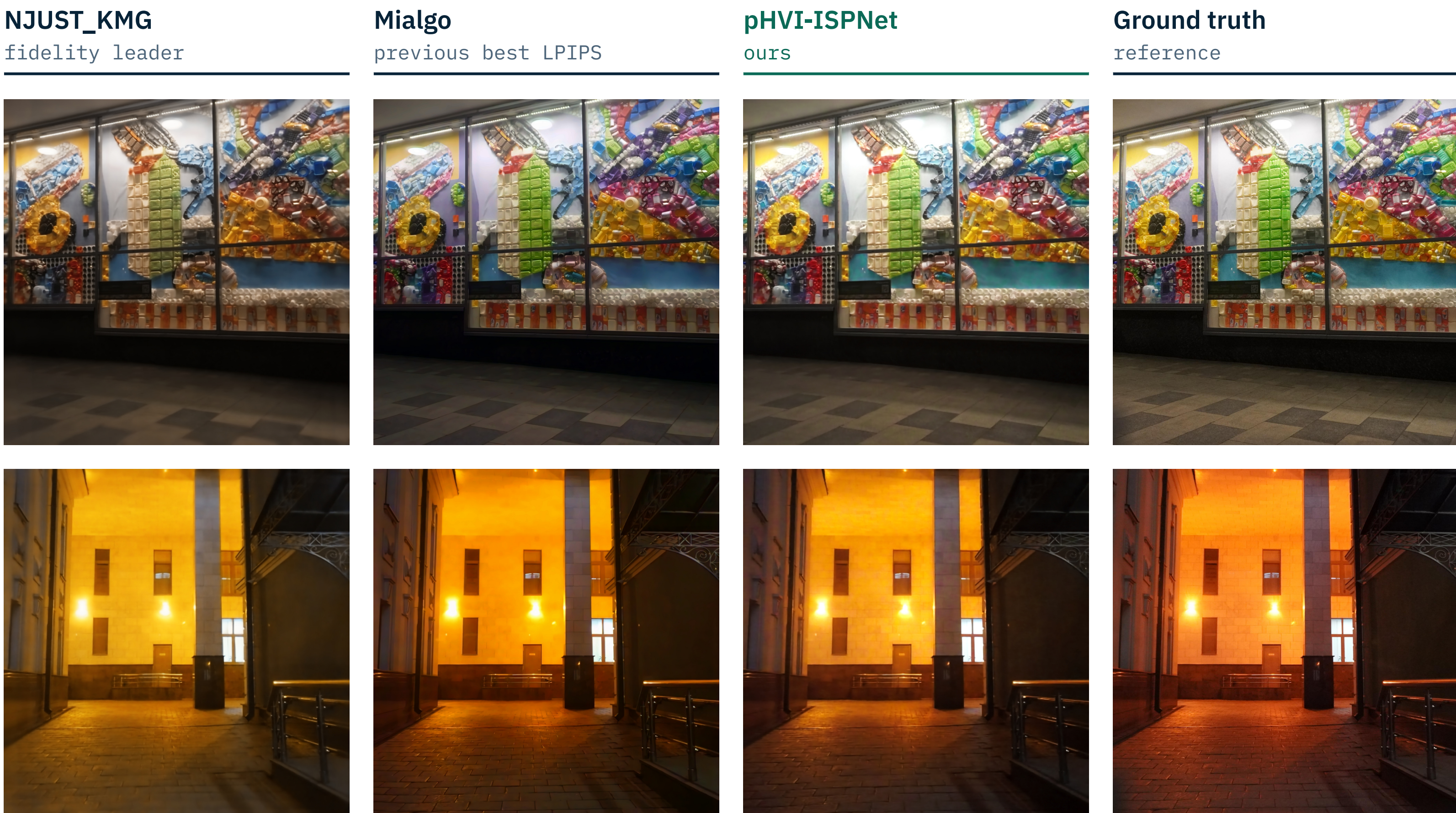
Dynamic loss weight α

A per-sample coefficient from the ratio of mean intensities penalizes both under- and over-exposure, stabilizing training across exposures.

ΔE 5.53 → 5.42
PSNR 23.58 → 23.63
LPIPS 0.379 → 0.388

3 Results

NTIRE 2025 night photography rendering benchmark



Full-resolution test scenes; no input column, since the RAW frame is effectively black before rendering. Bottom row: fidelity-tuned output turns yellow-green against a red-brown reference.

Method	PSNR ↑	SSIM ↑	ΔE ↓	LPIPS ↓
OzU-VGL	18.93	0.677	11.94	0.586
colab	19.15	0.635	10.27	0.476
AITH_ITMO	20.58	0.718	–	–
POLYU-AISP	21.23	0.737	7.53	0.463
psykhexx	21.70	0.754	7.48	0.455
DnF	23.06	0.781	6.41	0.454
NoTeam	21.22	0.699	8.62	0.453
PSU	22.21	0.758	7.04	0.441
NJUST_KMG	23.82	0.793	5.85	0.433
sorange	21.16	0.744	7.82	0.429
Mialgo	23.06	0.784	6.52	0.400
pHVI-ISPNet	23.63	0.785	5.42	0.388

Sorted by LPIPS, bold marks the best score in a column. Ours takes both perceptual columns and sits second on both fidelity columns.

Take-away

Decoupling color from intensity, carrying detail through the network with wavelets, and supervising color at the level of feature distributions buys **state-of-the-art perceptual quality** – ΔE 5.42 and LPIPS 0.388 – at no meaningful cost in fidelity: 23.63 dB and SSIM 0.785, within 0.19 dB of the challenge's fidelity leader.



read the paper

Dataset NTIRE 2025 NPR, Huawei RAW to Sony RGB	Training 768 × 768 crops, 2500 epochs, batch 2	Optimizer AdamW 2×10^{-4} , cosine to 1×10^{-5}	Hardware 2 × RTX 2080 Ti, PyTorch	Paper doi 10.1109/ICIP61757.2026.11630277
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